

← DIMENSION MISMATCH →

CRL

Within CRL

DGP COMPLEXITY →

Beyond CRL

Independent

Correlated

Functionally
constrained

(D1)

(D2)

(D3 / D4)

Overcomplete

$m > d$

(E5–E8)

FN: distributed codes

MCC-S

MCC-P, DCI-D, R^2

FN: correlation +
distributed codes

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Most metrics

FN: redundancy +
distribution compound

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All coordinate-wise

Matched

$m = d$

(E1–E3)

Metrics calibrated

MCC-P, MCC-S, R^2 , DCI-D

MIG

FN: correlation
confound

MCC-P, MCC-S, R^2

DCI-D, MIG, InfoM

FN: predictability
conflation

MCC-P, MCC-S

DCI-D, R^2 , MIG

Undercomplete

$m < d$

(E4)

FN: partial recovery
penalised

MCC-P, MCC-S, R^2

DCI-D, MIG

FN: partial recovery +
correlation confound

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Most metrics

FN: lossless vs lossy
indistinguishable

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All metrics

Null
baseline

E10 (Uniform noise)

Valid null (≈ 0)

Most metrics pass



Calibrated



Some metrics fail



Most metrics fail



All metrics fail